Customer Transaction Prediction - Project Summary

Project Objective
Develop a machine learning model to predict whether a customer will make a transaction based on anonymized customer features.

Dataset Overview

Dataset contains 200,000 records and 202 columns. Features include an ID column, a target variable, and 200 anonymized numerical variables (var_0 to var_199).

Data Preprocessing

Checked for missing values and duplicate records. Features were separated from the target variable, and the data was split into training (80%) and testing (20%) sets.

Models Implemented

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier
4. Tuned Random Forest using GridSearchCV

Model Performance

Logistic Regression Accuracy: 91.00%
Decision Tree Accuracy: 83.43%
Random Forest Accuracy: 89.76%
Tuned Random Forest Accuracy: 89.76%
ROC-AUC (Tuned Random Forest): 0.8203

Key Findings

Logistic Regression achieved the highest overall accuracy. Decision Tree showed signs of overfitting with 100% training accuracy but lower test accuracy. Random Forest improved stability but struggled to identify positive-class transactions due to class imbalance.

Feature Importance

Random Forest feature importance analysis identified variables such as var_81, var_12, var_139, var_53, and var_110 among the most influential predictors.

Business Conclusion

The project successfully predicts customer transaction behavior. Logistic Regression is the recommended model because it delivered the best test accuracy and more balanced classification performance compared with tree-based models.

Future Improvements

Address class imbalance using SMOTE or class weighting, perform advanced feature engineering, optimize hyperparameters further, and evaluate boosting algorithms such as XGBoost, LightGBM, or CatBoost.